Asymmetry effects in untrusted noise securuty analysis

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1 Imperfect gaussian measurements

We begin with a brief discussion of asymmetrical coherent measurements, schematically depicted in Fig. 1. Asymmetry effects are induced by imbalance (unequal transmission and reflection coefficients) of the beamsplitters and unequal quantum efficiency of the detectors. For explicit derivations of the expressions in this section see Ref. [1].

Q symbol of the POVM describing noisy homodyne measurements (see Fig. 1a) is given by

$$\mathsf{P}_G^{(\mathrm{H})}(x) = \frac{1}{\sqrt{2\pi\sigma_G}} \exp\left[-\frac{(x - \langle \hat{x}_\phi \rangle)^2}{2\sigma_x}\right], \tag{1}$$

$$\langle \hat{x}_\phi \rangle \equiv \langle \alpha | \hat{x}_\phi | \alpha \rangle = 2\,\mathrm{Re}\,\alpha e^{-i\phi}, \quad \phi = \arg \alpha_L, \tag{2}$$

$$\hat{x}_\phi = \hat{a} e^{-i\phi} + \hat{a}^\dagger e^{i\phi} \tag{3}$$

where α (α_L) is the complex amplitude of signal (LO) mode, $\hat{x}_\phi$ is the phase-rotated quadrature operator; x is the quadrature variable

$$x \equiv \frac{\mu}{(\eta_1 + \eta_2) CS\, |\alpha_L|} - \frac{\eta_1 S^2 - \eta_2 C^2}{(\eta_1 + \eta_2)\, CS} |\alpha_L| \tag{4}$$

and σ_x is the quadrature variance

$$\sigma_x \equiv \frac{\eta_1 S^2 + \eta_2 C^2}{[(\eta_1 + \eta_2) CS]^2}. \tag{5}$$

It is known that POVMs of noisy measurements can be introduced using the dual of a quantum channel acting on POVMs representing perfect (noiseless) measurements [2]. From the Q symbol (1), using formula for overlap of two Gaussian states [2], we deduce the following expression for the covariance matrix of noisy homodyne measurements

$$\Sigma_m^{(\mathrm{H})} = \lim_{r \to +\infty} R(\phi)\, \mathrm{diag}(e^{-2r} + \sigma_N, e^{2r})\, R^{\mathrm{T}}(\phi) = \Sigma_{\mathrm{id}}^{(\mathrm{H})} + \Sigma_N^{(\mathrm{H})}, \tag{6}$$

$$\Sigma_{\mathrm{id}}^{(\mathrm{H})} = \lim_{r \to +\infty} R(\phi)\, \mathrm{diag}(e^{-2r}, e^{2r})\, R^{\mathrm{T}}(\phi) \tag{7}$$

$$\Sigma_N^{(\mathrm{H})} = R(\phi)\, \mathrm{diag}(\sigma_N, 0)\, R^{\mathrm{T}}(\phi), \tag{8}$$

where $R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$ is the rotation matrix; σ_N is the excess noise variance that accounts for asymmetry effects is given by

$$\sigma_N = \sigma_x - 1. \tag{9}$$

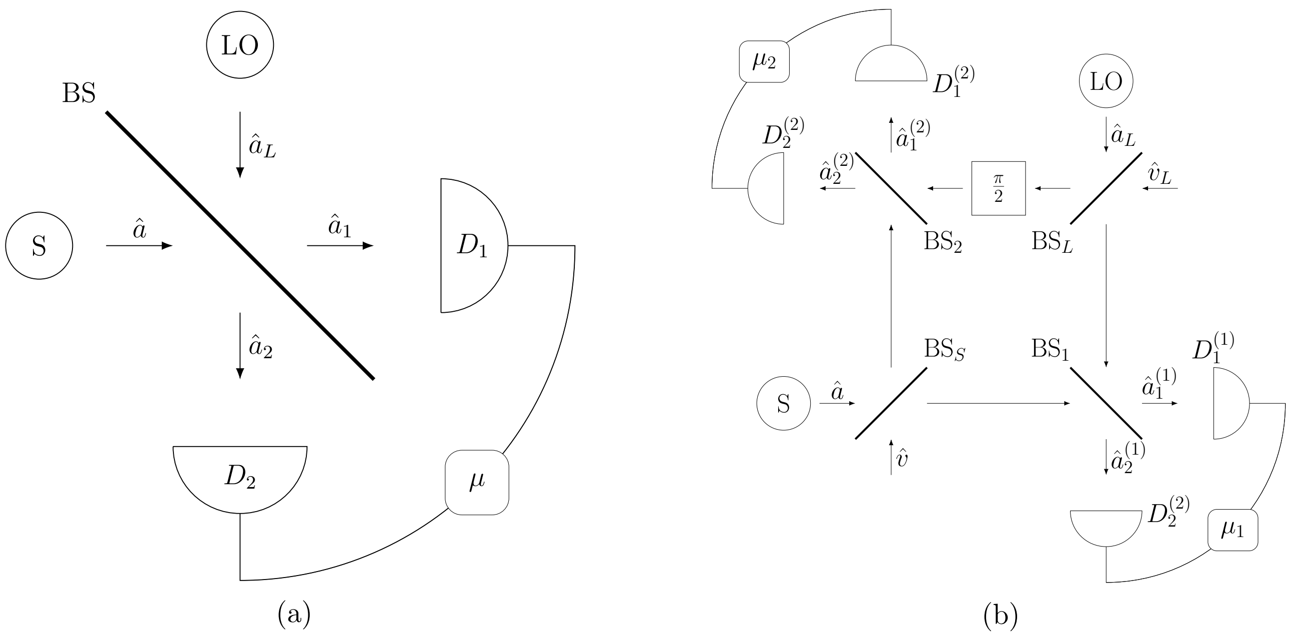


Figure 1: (a) Scheme of homodyne receiver. S (LO) is the source of signal (reference) mode, BS is the beamsplitter with transmission and reflection coefficients C and S, respectively; photodetectors D_1 and D_2 have quantum efficiencies η_1 and η_2, and $\mu = m_1 - m_2$ is the photon count difference. (b) Scheme of double homodyne receiver. S (LO) is the source of signal (reference) mode, BS_S (BS_L) is the signal mode (local oscillator) beam splitter; $\frac{\pi}{2}$ is the quarter wave phase shifter; BS_i is the beam splitter of ith homodyne; $D_{1,2}^{(i)}$ are the photodetectors of the ith homodyne; and $\mu_i = m_1^{(i)} - m_2^{(i)}$ is the photon count difference registered by the detectors of ith homodyne.

For double homodyne detection, Q symbol is given by

$$\mathsf{P}_G^{(\mathrm{DH})}(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}} \exp\left[-\frac{(x_1 - \operatorname{Re}\alpha e^{-i\phi})^2}{\sigma_1} - \frac{(x_2 - \operatorname{Im}\alpha e^{-i\phi})^2}{\sigma_2}\right], \tag{10}$$

$$\sigma_G^{(i)} = (\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2)|\alpha_L^{(i)}|^2, \quad \alpha_L^{(1)} = C_L \alpha_L, \quad \alpha_L^{(2)} = -i S_L \alpha_L, \tag{11}$$

where x_i are the quadrature variables given by

$$x_1 = \frac{1}{2(\eta_1^{(1)} + \eta_2^{(1)}) C_1 S_1 C_S} \left\{ \frac{\mu_1}{|\alpha_L^{(1)}|} - (\eta_1^{(1)} S_1^2 - \eta_2^{(1)} C_1^2)|\alpha_L^{(1)}| \right\}, \tag{12}$$

$$x_2 = \frac{1}{2(\eta_1^{(2)} + \eta_2^{(2)}) C_2 S_2 S_S} \left\{ \frac{\mu_2}{|\alpha_L^{(2)}|} - (\eta_1^{(2)} S_2^2 - \eta_2^{(2)} C_2^2)|\alpha_L^{(2)}| \right\}, \tag{13}$$

and relations

$$\sigma_1 = \frac{\sigma_x^{(1)}}{2C_S^2}, \quad \sigma_2 = \frac{\sigma_x^{(2)}}{2S_S^2}, \tag{14}$$

$$\sigma_x^{(i)} = \frac{\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2}{\left[(\eta_1^{(i)} + \eta_2^{(i)}) C_i S_i\right]^2} \tag{15}$$

give the quadrature variances σ_1 and σ_2.

The covariance matrix of the POVM reads

$$\Sigma_m^{(\mathrm{DH})} = R(\phi)\,\mathrm{diag}(\delta_1, \delta_2) R^{\mathrm{T}}(\phi), \tag{16}$$
$$\delta_i = 2\sigma_i - 1. \tag{17}$$

A distinctive feature of double homodyne detection is that the POVM describing sharp measurements uses the set of squeezed states, with covariance matrices given by

$$\Sigma_{\mathrm{id}}^{(\mathrm{DH})} = R(\phi) \begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix} R^{\mathrm{T}}(\phi), \tag{18}$$

so that excess asymmetry noise covariance matrix

$$\Sigma_N^{(\mathrm{DH})}(r) = \Sigma_m^{(\mathrm{DH})} - \Sigma_{\mathrm{id}}^{(\mathrm{DH})} = R(\phi) \begin{pmatrix} 4\sigma_N^{(1)}(r) & 0 \\ 0 & 4\sigma_N^{(2)}(r) \end{pmatrix} R^{\mathrm{T}}(\phi) \geq 0, \tag{19}$$
$$4\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \quad 4\sigma_N^{(2)}(r) = \delta_2 - e^{-2r}, \tag{20}$$

is explicitly dependent on squeezing parameter that lies in the interval defined by measurement asymmetry:

$$r \in [r_2, r_1]\,, \ r_1 = \ln \delta_1^{1/2}, \ r_2 = \ln \delta_2^{-1/2}. \tag{21}$$

2 Gaussians channels

In what follows, we shall use the attenuator quantum channel [2], characterized by matrices

$$X^{(\gamma)} = \cos\theta^{(\gamma)} \mathbb{1}, \quad Y^{(\gamma)} = \sin^2\theta^{(\gamma)} \Sigma_{\mathrm{aux}}^{(\gamma)}, \tag{22}$$

$$\mathcal{E}_N^{(\gamma)} : \hat{\Pi}_{\mathrm{id}} \mapsto \hat{\Pi}_m^{(\gamma)} = \mathcal{E}_N^{*(\gamma)}(\hat{\Pi}_{\mathrm{id}}) = \frac{2}{\pi \cos\theta^{(\gamma)} \sin\theta^{(\gamma)} \sqrt{\det \Sigma_{\mathrm{aux}}^{(\gamma)}}} \int_{\mathbb{C}} \mathrm{d}^2\beta \, \exp\left\{-2\mathbf{s}^T \Sigma_{\mathrm{aux}}^{-1} \mathbf{s}\right\} \hat{D}(\cos\theta^{(\gamma)}\,\beta)\, \hat{\Pi}_{\mathrm{id}}\, \hat{D}(-\cos\theta^{(\gamma)}\,\beta). \tag{23}$$

where $\mathbf{s}^{\mathrm{T}} = (\cos\theta^{(\gamma)}\,\mathrm{Re}(\beta), \cos\theta^{(\gamma)}\,\mathrm{Im}(\beta))$, $\mathrm{d}^2\beta = \mathrm{d}\beta_1 \mathrm{d}\beta_2$ and $\hat{D}(\beta)$ is the displacement operator given by

$$\hat{D}(\beta) = e^{\beta \hat{a}^\dagger - \beta^* \hat{a}}. \tag{24}$$

Dual of (23), $\mathcal{E}_N^{*(\gamma)}$, transforms the POVM of the ideal measurement into the POVM of the noisy measurements by changing respective covariance matrix $\Sigma_{\mathrm{id}}^{(\gamma)}$ into $\Sigma_m^{(\gamma)}$ as follows:

$$\mathcal{E}_N^{*(\gamma)} : \quad \Sigma_{\mathrm{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \frac{1}{\cos^2\theta^{(\gamma)}} \Sigma_{\mathrm{id}}^{(\gamma)} + \tan^2\theta^{(\gamma)} \Sigma_{\mathrm{aux}}^{(\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}. \tag{25}$$

Now we derive explicit experessions coefficients of (25) (see Eq. (22)).

For homodyne detection, since covariance matrices of ideal and noisy measurements (see Eq. (7) and (8)) are taking limits, implying that

$$\frac{1}{\cos^2\theta} \Sigma_{\mathrm{id}}^{(\mathrm{H})} \sim \Sigma_{\mathrm{id}}^{(\mathrm{H})}, \quad e^{2r} + \mathrm{const} \sim e^{2r}, \tag{26}$$

we may write the effective excess noise covariance matrix

$$\Sigma_N^{\mathrm{(H)\,eff}} = R(\phi)\,\mathrm{diag}(\sigma_N, \sigma_N)\,R^{\mathrm{T}}(\phi), \tag{27}$$

so that enviroment is decribed by vacuum, i.e.

$$\Sigma_{\mathrm{aux}}^{\mathrm{(H)}} = \mathbb{1}. \tag{28}$$

Solving for $\theta^{\mathrm{(H)}}$ yields

$$\cos\theta^{\mathrm{(H)}} = \sqrt{\frac{1}{\sigma_x}}, \quad \sin\theta^{\mathrm{(H)}} = \sqrt{\frac{\sigma_N}{\sigma_x}}. \tag{29}$$

For double homodyne detection, first, we note that, since $\Sigma_m^{\mathrm{(DH)}}$ is anisotropic, as generally quadrature variances $\delta_{1,2}$ are not equal, $\delta_1 \neq \delta_2$, ancilla state must be anisotropic as well. Without loss of generality, we will consider the ancilla to be in pure state with $\det \Sigma_{\mathrm{aux}}^{\mathrm{(DH)}} = 1$,

$$\Sigma_{\mathrm{aux}}^{\mathrm{(DH)}} = R(\phi)\,\mathrm{diag}\left(a, \frac{1}{a}\right)\,R^{\mathrm{T}}(\phi), \quad a > 0. \tag{30}$$

From Eq. (25), we arrive at the expression for $\cos^2\theta^{\mathrm{(DH)}}$ in the form

$$\cos^2\theta^{\mathrm{(DH)}}(r) = \frac{e^{2r} + a(r)}{\delta_1 + a(r)}, \tag{31}$$

$$a(r) = \frac{1}{2}\left[-b + \sqrt{b^2 - 4c}\right], \quad b = \frac{-\delta_1 e^{-2r} + \delta_2 e^{2r}}{4\sigma_N^{(2)}}, \quad c = \frac{-\sigma_N^{(1)}}{\sigma_N^{(2)}}. \tag{32}$$

It's worth noting that beyond the r interval ancilla becomes strictly unphysical, i.e. $a \in (0, +\infty) \iff r \in (r_2, r_1)$. This is due to $\sigma_N^{(i)}$ becoming negative, so that the argument of the square root in Eq. (32) becomes negative as well.

We will also consider classical mixing (additive noise) channel [2] used in Ref. [1]:

$$\Phi_N : \hat{\Pi}_{\mathrm{id}} \mapsto \hat{\Pi}_G = \Phi_N(\hat{\Pi}_{\mathrm{id}}) = \frac{2}{\pi\sqrt{\det \Sigma_N}}$$
$$\times \int_{\mathbb{C}} \mathrm{d}^2\beta \exp\{-2\mathbf{s}^T \Sigma_N^{-1}\mathbf{s}\}\hat{D}(\beta)\hat{\Pi}_{\mathrm{id}}\hat{D}(-\beta), \tag{33}$$
$$\mathbf{s}^{\mathrm{T}} = (\mathrm{Re}(\beta), \mathrm{Im}(\beta)) = (\beta_1, \beta_2), \quad \mathrm{d}^2\beta = \mathrm{d}\beta_1\mathrm{d}\beta_2. \tag{34}$$

This is a self-dual channel, $\Phi_N^* = \Phi_N$, that transforms the POVM of the ideal projective Gaussian measurements, $\hat{\Pi}_{\mathrm{id}}$, into the POVM of the noisy measurements, $\hat{\Pi}_G$, by changing the covariance matrix Σ_{id} as follows

$$\Phi_N : \Sigma_{\mathrm{id}} \mapsto \Sigma_G = \Sigma_{\mathrm{id}} + \Sigma_N. \tag{35}$$

3 GMCS protocol

In the GMCS protocol, Alice sends coherent states $|\alpha = \alpha_1 + i\alpha_2\rangle$ drawn from the Gaussian prior

$$p_A(\alpha) = \frac{1}{\pi V_{\mathrm{A}}} \exp\left[-\frac{|\alpha|^2}{2V_{\mathrm{A}}}\right]. \tag{36}$$

The states pass the composite Gaussian channel $\mathcal{N}_{A\to B} = \Phi_{\xi_{\rm ch}} \circ \mathcal{E}_{T_{\rm ch}}$, where $\mathcal{E}_{T_{\rm ch}}$ is pure loss channel with transmittance $T_{\rm ch}$ and Φ_ξ adds excess noise with covariance $\Sigma_{\xi_{\rm ch}} = \xi_{\rm ch}\mathbb{1}$. Then, the receiver uses either homodyne or double homodyne detection to measure the states. Mutual information between legitimate parties in each case is then given by the expressions (see Ref. [1] for derivation):

$$I_{\rm AB}^{(\rm H)} = \frac{1}{2}\log\left[1 + \frac{4T_{\rm ch}V_{\rm A}}{\sigma_x + \xi_{\rm ch}}\right], \tag{37}$$

$$I_{\rm AB}^{(\rm DH)} = \frac{1}{2}\sum_i \log\left[1 + \frac{4T_{\rm ch}V_{\rm A}}{2\sigma_i + \xi_{\rm ch}}\right], \tag{38}$$

where σ_x and σ_i given by Eqs. (5) and (15), respectively.

The covariance matrix of TMSV state with the Bob's mode transmitted through the noisy channel (see Ref. [3])

$$\mathcal{I}_A \otimes \Phi_{\xi_{\rm ch}} \circ \mathcal{E}_{T_{\rm ch}} : \quad \Sigma_{\rm TMSV} = \begin{pmatrix} V\mathbb{1} & \sqrt{V^2-1}\sigma_z \\ \sqrt{V^2-1}\sigma_z & V\mathbb{1} \end{pmatrix} \mapsto \Sigma_{\rm AB} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\rm B}\mathbb{1} \end{pmatrix}, \tag{39}$$

where $\mathcal{I}_A$ is the identity channel acting on the subsystem A and the parameters of the covariance matrix $\Sigma_{\rm AB}$ are given by

$$V = 1 + 4V_{\rm A}, \quad c = \sqrt{T_{\rm ch}(V^2-1)}, \tag{40}$$

$$V_{\rm B} = T_{\rm ch}(V-1) + 1 + \xi_{\rm ch}. \tag{41}$$

In untrusted noise scenario, where excess noise caused by asymmetry effects is assumed to be in control of Eve, channel that models the excess noise can be modeled in two equivalent ways, per Eq. (25) and Eq. (33).

We consider local action of either attenuator channel $\mathcal{E}_N^{(\gamma)}$ (25) or additive noise channel $\Psi^{(\gamma)}$ (33), on second mode of TMSVS shared by Alice and Bob (39). In both cases, mutual information is given by $I_{\rm AB}^{(\gamma)}$ (37), (37), as it is classical.

It follows

$$\mathcal{I}_A \otimes \mathcal{E}_N^{(\gamma)} : \Sigma_{\rm AB} \mapsto \Sigma_{\rm AB}^{(\gamma;\mathcal{E}_N)} = \begin{pmatrix} V\mathbb{1} & c\cos\theta^{(\gamma)}\sigma_z \\ c\cos\theta^{(\gamma)}\sigma_z & \cos^2\theta^{(\gamma)}\left(V_{\rm B} + \tan^2\theta^{(\gamma)}\Sigma_{\rm aux}^{(\gamma)}\right) \end{pmatrix}, \tag{42}$$

$$\mathcal{I}_A \otimes \Phi_N^{(\gamma)} : \Sigma_{\rm AB} \mapsto \Sigma_{\rm AB}^{(\gamma;\Phi)} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\rm B}\mathbb{1} + \Sigma_N^{(\gamma)} \end{pmatrix}, \quad \gamma \in \{\rm H, DH\}. \tag{43}$$

When the channel $C \in \{\Phi_N^{(\gamma)}, \mathcal{E}_N^{(\gamma)}\}$ is under control of the eavesdropper, Eve holds a purification of the state $\hat{\rho}_{\rm AB}^{(\gamma;C)} = \mathcal{I}_A \otimes C^{(\gamma)}(\hat{\rho}_{\rm AB})$ and the measurements performed by the receiver are assumed to be noiseless. In this scenario, the difference between the joint and conditional entropies

$$\chi_{\rm EB}^{(\gamma;C)} = S_{\rm E} - S_{\rm E|B} = S_{\rm AB}^{(\gamma;C)} - S_{\rm A|B}^{(\gamma;C)}, \quad C \in \{\Phi_N^{(\gamma)}, \mathcal{E}_N^{(\gamma)}\}. \tag{44}$$

gives the Holevo information. The joint entropy

$$S_{\rm AB}^{(\gamma;C)} = g(\nu_1^{(\gamma;C)}) + g(\nu_2^{(\gamma;C)}), \tag{45}$$

$$g(\nu) = \frac{\nu+1}{2}\log\frac{\nu+1}{2} - \frac{\nu-1}{2}\log\frac{\nu-1}{2} \tag{46}$$

then can be expressed in terms of the symplectic eigenvalues of the covariance matrix (43) given by

$$\nu_{1,2}^{(\gamma;C)} = \sqrt{\Delta_{\gamma;C}/2 \pm \sqrt{\Delta^2_{\gamma;C}/4 - D_{\gamma;C}}}, \quad D_{\gamma;C} = \det \Sigma_{\mathrm{AB}}^{(\gamma;C)}, \tag{47}$$

$$\Delta_{\gamma;C} = \det([\Sigma_{\mathrm{AB}}^{(\gamma;C)}]_{11}) + \det([\Sigma_{\mathrm{AB}}^{(\gamma;C)}]_{22}) + 2\det([\Sigma_{\mathrm{AB}}^{(\gamma;C)}]_{12}), \tag{48}$$

where $[\Sigma_{\mathrm{AB}}^{(\gamma;C)}]_{ii}$ ($[\Sigma_{\mathrm{AB}}^{(\gamma;C)}]_{ij}$) are diagonal (off-diagonal) blocks of $\Sigma_{\mathrm{AB}}^{(\gamma;C)}$.

Conditional entropies $S_{\mathrm{E|B}} = S_{\mathrm{A|B}}^{(\gamma)}$ are calculated from respective conditional covariance matrices,

$$S_{\mathrm{A|B}}^{(\gamma)} = g(\nu_3^{(\gamma)}), \quad \nu_3^{(\gamma)} = \sqrt{\det \Sigma_{\mathrm{A|B}}^{(\gamma)}}, \tag{49}$$

$$\Sigma_{\mathrm{A|B}}^{(\gamma)} = V\mathbb{1} - c^2\sigma_z(V_{\mathrm{B}}\mathbb{1} + \Sigma_m^{(\gamma)})^{-1}\sigma_z. \tag{50}$$

Direct computations show that these conditional matrices and therefore their symplectic eigenvalues are independent on channel model. Symplectic eigenvalues are given by:

$$\nu_3^{(\mathrm{H})} = \sqrt{V\left(V - \frac{c^2}{V_{\mathrm{B}} + \sigma_N}\right)}, \tag{51}$$

$$\nu_3^{(\mathrm{DH})} = V\sqrt{\frac{\left(V_{\mathrm{B}} + \delta_1 - \frac{c^2}{V}\right)\left(V_{\mathrm{B}} + \delta_2 - \frac{c^2}{V}\right)}{\left(V_{\mathrm{B}} + \delta_1\right)\left(V_{\mathrm{B}} + \delta_2\right)}}, \tag{52}$$

From mutual information and Holevo information evaluation of the asymptotic secret fraction is done per Devetak-Winter formula [4]

$$K^{(\gamma;C)} = \beta I_{AB}^{(\gamma;C)} - \chi_{EB}^{(\gamma;C)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}, C \in \{\Phi_N^{(\gamma)}, \mathcal{E}_N^{(\gamma)}\}. \tag{53}$$

As symplectic eigenvalues for additive noise channel are calculated per formulas Eq. (47) and do not allow for simplification, in the following sections we focus only on analytic expressions for symplectic eigenvalues for attenuator channel model.

4 Homodyne detection

For attenuator channel, from Eq. (42) and (29) we have

$$\Sigma_{\mathrm{AB}}^{(\mathrm{H};\mathcal{E}_N)} = \begin{pmatrix} V\mathbb{1} & \frac{c}{\sqrt{\sigma_x}}\sigma_z \\ \frac{c}{\sqrt{\sigma_x}}\sigma_z & \frac{1}{\sigma_x}\left(V_{\mathrm{B}} + \sigma_N\right)\mathbb{1} \end{pmatrix} \equiv \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_{\mathrm{B}}\mathbb{1} \end{pmatrix}, \tag{54}$$

$$\tilde{V}_{\mathrm{B}} \equiv T(V-1) + 1 + \xi, \tag{55}$$

$$\tilde{c} \equiv \sqrt{T(V^2-1)}, \tag{56}$$

where we defined scaled Bob's variance and scaled correlations between Alice and Bob $\tilde{V}_{\mathrm{B}}$ and $\tilde{c}$, using total channel transmission T that consists of channel transmission T_{ch} and detection losses $\frac{1}{\sigma_x}$, as well as scaled noise variance ξ:

$$T = T_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}, \tag{57}$$

$$\xi \equiv \xi_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}. \tag{58}$$

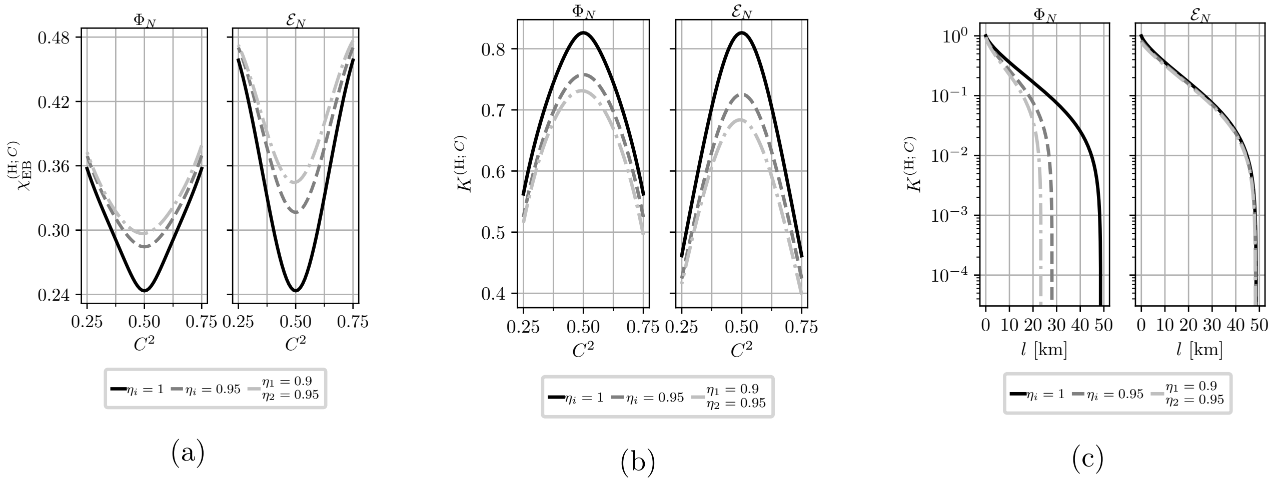


Figure 2: (a) Holevo information, $\chi_{\mathrm{EB}}^{(\mathrm{H};C)}$ and (b) asymptotic secret fraction, $K^{(\mathrm{H};C)}$, as a function of the beam splitter transmission, C^2, of the homodyne receiver computed at different values of the detector efficiencies for additive noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5; T_{\mathrm{ch}} = 0.95; \xi_{\mathrm{ch}} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(\mathrm{H})}$ versus transmission length l for additive noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; losses are 20 dB per 100 km.

It follows that for homodyne detection the effect of measurement asymmetry is equivalent to scaling both channel transmission (57) and channel noise (58) by a factor of σ_x^{-1}.

Symplectic eigenvalues of $\Sigma_{\mathrm{AB}}^{(\mathrm{H};\mathcal{E}_N)}$, $\nu_{1,2}^{(\mathrm{H};\mathcal{E}_N)}$, can be computed using formulas [5]:

$$\nu_{1,2}^{(\mathrm{H};\mathcal{E}_N)} = \frac{z \pm [\tilde{V}_{\mathrm{B}} - V]}{2}, \quad z = \sqrt{\left(V + \tilde{V}_{\mathrm{B}}\right)^2 - 4\tilde{c}^2}, \tag{59}$$

and then substitued in Eq. (45) to calculate joint entropy $S_{\mathrm{AB}}^{(\mathrm{H};\mathcal{E}_N)}$.

Fig. 2 illustrates the effects of homodyne measurement asymmetry on Holevo information (44) and asymptotic secret fraction (53) versus beam splitter transmission C^2, as well as secret fraction versus transmission length, for both additive noise (43) and attenuator (42) channels. While channel noise is reduced in this setup (58), so is channel transmission (57), and, since transmission is scaled by modulation variance $V \gg 1$ in every expression, as well as reduced mutual information with greater asymmetry per Eq. (37), attenuator channel has negative effect. As shown in Fig. 2a and 2b, the attenuator channel results in higher Holevo information, and thus lower secret fraction, at high transmission for fixed asymmetry parameters. However, Fig. 2c reveals that the attenuator channel achieves significantly greater reach (maximum transmission length at which a positive key rate is obtained) than the additive noise channel.

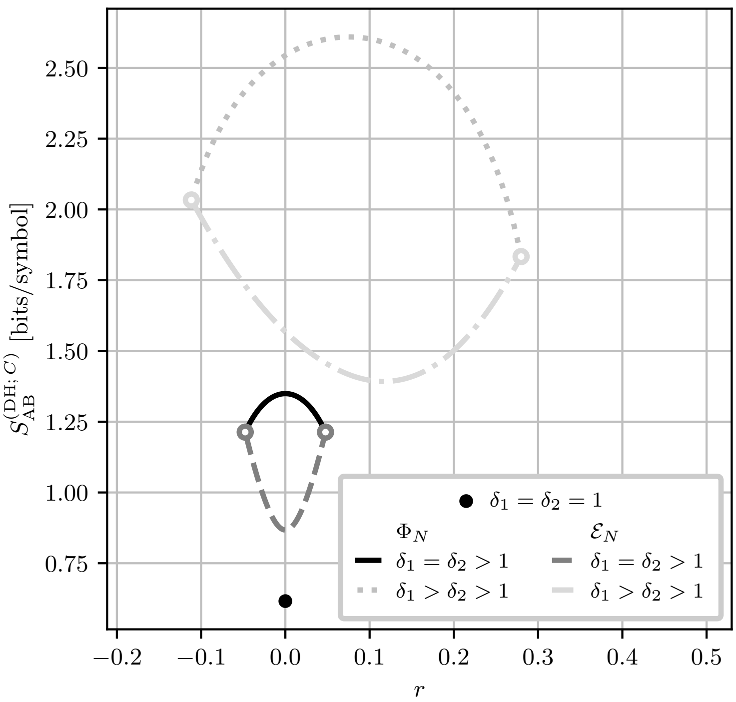


Figure 3: Joint entropy, $S_{\mathrm{AB}}^{(\mathrm{DH};C)}$, as a function of the squeezing parameter r at different values of $\delta_{1,2}$. Black dot corresponds to the case of perfect measurements. Solid and dashdotted curves correspond to additive noise channel model, while dashed and dotted curves are for attenuator channel model. Color-matched white-filled dots mark points where joint entropy is undefined at the boundaries of the r interval (21) (see Eq. (32) and main text). Values of $\delta_{1,2}$ are: $\delta_{1,2} = 1.1$ for solid and dashed curves, $\delta_1 = 1.75, \delta_2 = 1.25$ for dashdotted and dotted curves. The channel parameters are: $V = 5$; $T_{\mathrm{ch}} = 0.95$; and $\xi_{\mathrm{ch}} = 10^{-2}$.

5 Double homodyne

For attenuator, from Eq. (42) and (31) we have

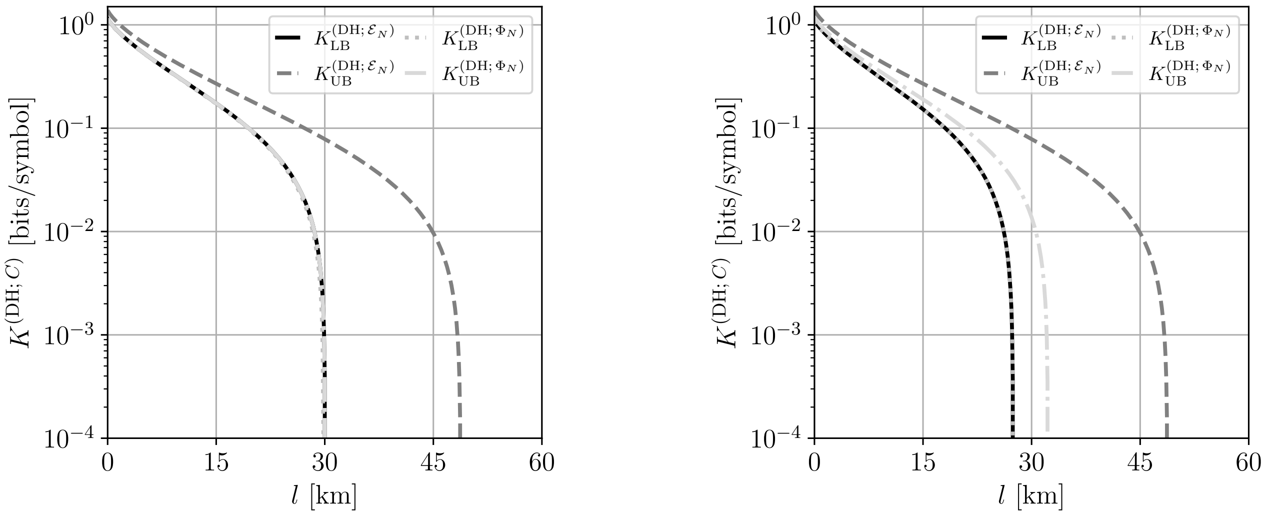
$$\Sigma_{\mathrm{AB}}^{(\mathrm{DH};\mathcal{E}_N)} = \begin{pmatrix} V\mathbb{1} & & c\cos\theta^{(\mathrm{DH};\mathcal{E}_N)}\sigma_z & \\ c\cos\theta^{(\mathrm{DH};\mathcal{E}_N)}\sigma_z & \begin{matrix}\cos^2\theta^{(\mathrm{DH};\mathcal{E}_N)}\left(V_{\mathrm{B}} + \delta_1 - \frac{e^{2r}}{\cos^2\theta^{(\mathrm{DH};\mathcal{E}_N)}}\right) & 0 \\ 0 & \cos^2\theta^{(\mathrm{DH};\mathcal{E}_N)}\left(V_{\mathrm{B}} + \delta_2 - \frac{e^{-2r}}{\cos^2\theta^{(\mathrm{DH};\mathcal{E}_N)}}\right)\end{matrix} \end{pmatrix}. \tag{60}$$

Symplectic eigenvalues of this matrix can be calculated using invariants similarly to Eq. (47). It is obvious that these symplectic eigenvalues, and thereby joint entropy, is explicitly depended on squeezing parameter r (21), like in the case of additive noise channel.

Fig. 3 illustrates dependence of the joint entropy on the squeezing parameter, for both additive noise and attenuator channels. Referring to Fig. 3, in the limiting case of ideal double homodyne with $\delta_1 = \delta_2 = 1$, only one value of $r = 0$ is allowed (see Ref. [1]), with matching joint entropies for both channels (black dot on plot).

When $\delta_1 = \delta_2 > 1$, dependence of joint entropy becomes symmetrical around 0 for both channels. For additive noise channel, $r = 0$ is the point of maximal joint entropy, while for attenuator channel $r = 0$ corresponds to minimal joint entropy. Moreover, while joint entropy is unidefined at $r = r_{2,1}$ (see Eq. (31)), the values tend to the same values of joint entropy in the case of additive noise channel, as signified by color-matched white-filled dots in Fig. 3.

In asymmetrical case where $\delta_1 \neq \delta_2$, joint entropies for both channel models become asymmetrical relative to their extrema, resulting in unequal joint entropies at $r = r_{2,1}$. For additive noise channel, joint entropy has a distintive maxima, while for attenuator channel joint entropy has a distintive minima.



(a) $q = 1$

(b) $q = 1.25$

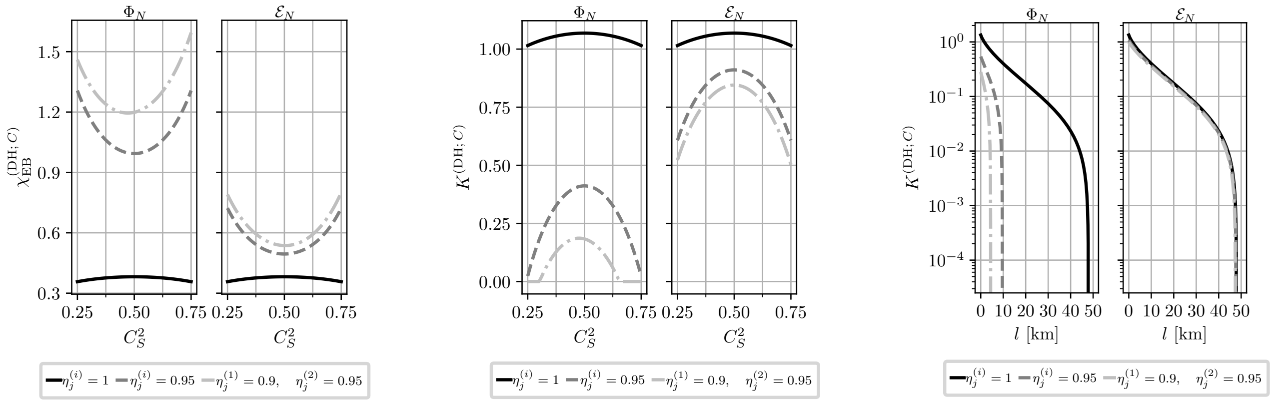
Figure 4: Asymptotic secret fraction $K^{(\mathrm{DH};C)}$ versus channel length l, computed for fixed $\sigma_x^{(i)} = 1.05$ (see Eq.(15)) and various imbalance ratios $q \equiv \frac{S_S^2}{C_S^2}$ for additive noise and attenuator channel models (see Fig. 3). The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; losses are 20 dB per 100 km.

Fig. 4 illustrates lower and upper bounds of the asymptotic secret fraction versus transmission length, for both additive noise (43) and attenuator (42) channels. Note that for chosen $\sigma_x^{(i)}$ lower bounds for both channels coincide, because for $\sigma_x^{(i)} \approx 1$ maxima of joint entropy for both models is either at $r = r_1$ or $r = r_2$. Morverover, the upper bound for additive channel in Fig. 4a is very close (0.2 km difference) to the lower bound.

As seen from Fig. 4, difference between lower and upper bound of the asymptotic secret fraction is greater in

Fig. 5 illustrates the dependence of Holevo information (44) and asymptotic secret fraction (53) on the signal beam splitter transmission C_S^2, as well as secret fraction versus transmission length if the receiver uses double homodyne detection, for both additive noise (43) and attenuator (42) channels. Unlike the homodyne detection case (see Fig. 2), Holevo information is significantly lower for attenuator channel, except for the ideal measurements case ($\eta_{1,2}^{(1,2)} = 1$ and whole C_S^2 axis, unlike in homodyne case where sharp measurements are strictly $\eta_{1,2} = 1$ and $C^2 = 0.5$) as both channels have no effect and thereby have equal joint entropies (black curve). High difference in Holevo information is due to the dependence of joint entropy on r (see Fig. 3) and fixing $r = \frac{r_2+r_1}{2}$ for both curves, effectively minimizing Holevo information for attenuator channel and maximizing it for additive noise channel.

Analogously to homodyne detection case (see Fig. 2c), in double homodyne detection case the attenuator channel has greater reach than the additive noise channel, as illustated by Fig. 5c.



(a) (b) (c)

Figure 5: (a) Holevo information, $\chi_{\mathrm{EB}}^{(\mathrm{DH});C}$ and (b) asymptotic secret fraction, $K^{(\mathrm{DH};C)}$, as a function of the signal beam splitter transmission, C_S^2, of the double homodyne receiver computed at different values of the detector efficiencies for additive noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5; T_{\mathrm{ch}} = 0.95; \xi_{\mathrm{ch}} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(\mathrm{DH};C)}$ versus transmission length l for additive noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; losses are 20 dB per 100 km.. For both channel models $r = \frac{r_2+r_1}{2}$ (see main text).

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